



Introduction

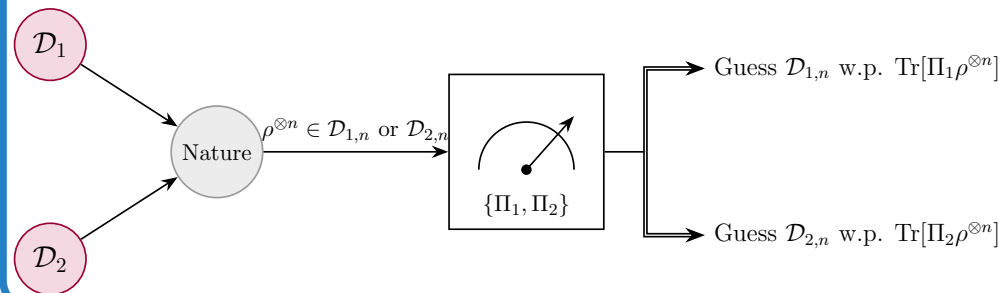
- We are given n copies of an unknown d -dimensional quantum state $\rho^{\otimes n}$, promised that ρ belongs to one of two **compact** sets $\mathcal{D}_1$ or $\mathcal{D}_2$.
- We assume the copies are i.i.d., so the hypotheses are the n -copy sets

$$\mathcal{D}_{1,n} := \{\rho_1^{\otimes n} : \rho_1 \in \mathcal{D}_1\} \text{ and } \mathcal{D}_{2,n} := \{\rho_2^{\otimes n} : \rho_2 \in \mathcal{D}_2\}.$$

- Composite QHT** addresses:

Which set does ρ belong to?

- To decide, we perform a binary POVM $\{\Pi_1, \Pi_2\}$, where outcome i favours $\mathcal{D}_{i,n}$.



Our Setting & Contributions

- We study the **sample complexity**.

Definition of Composite Sample Complexity

For $\delta \in (0, 1)$, the **composite sample complexity** is,

$$n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) := \inf\{n \in \mathbb{N} : P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) \leq \delta\}, \quad (1)$$

where $P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n})$ is the **minimum error probability**:

$$P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) = \inf_{0 \leq \Pi \leq I} \sup_{\rho_i \in \mathcal{D}_i} [p \text{Tr}[(I - \Pi)\rho_1^{\otimes n}] + (1-p) \text{Tr}[\Pi\rho_2^{\otimes n}]],$$

and $p \in (0, 1)$ is the prior on $\mathcal{D}_1$.

- So far, sample complexity has only been characterized in simple QHT:

Known Results: Simple QHT [1]

The following bounds hold:

$$\frac{\ln\left(\frac{p(1-p)}{\delta}\right)}{-2 \ln F(\rho_1, \rho_2)} \leq n^*(p, \rho_1, \rho_2, \delta) \leq \left[\inf_{s \in [0,1]} \frac{\ln\left(\frac{p^s(1-p)^{1-s}}{\delta}\right)}{-\ln \text{Tr}(\rho_1^s \rho_2^{1-s})} \right]. \quad (2)$$

- Our paper establishes the following.

- A lower bound matching the above for singleton sets.
- Upper bounds when one set is a single pure state, and for finite and infinite cardinality uncertainty sets.
- Sample complexity bounds for Private Composite QHT.

The Main Results

- Our results are expressed in terms of the **maximum fidelity**:

$$F_{\max} := \sup_{\rho_1 \in \mathcal{D}_1, \rho_2 \in \mathcal{D}_2} \|\sqrt{\rho_1} \sqrt{\rho_2}\|_1.$$

Theorem 1: Lower bound

Let $\mathcal{D}_1, \mathcal{D}_2$ be compact sets of d -dimensional quantum states. Then, for $p, \delta \in (0, 1)$ the following bound holds:

$$\frac{\ln\left(\frac{p(1-p)}{\delta}\right)}{-2 \ln F_{\max}} \leq n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta). \quad (3)$$

- Quantum state verification problem (QSVP): $\mathcal{D}_1 = \{|\psi_1\rangle\langle\psi_1|\}$, $\mathcal{D}_2$ is a compact set.

Theorem 2: Upper bound (QSVP)

Let $\mathcal{D}_1 := \{|\psi_1\rangle\langle\psi_1|\}$ and $\mathcal{D}_2$ be a compact set of d -dimensional quantum states. Then, for $p, \delta \in (0, 1)$ the following bound holds:

$$n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) \leq \left\lceil \frac{\ln\left(\frac{1-p}{\delta}\right)}{-\ln \sup_{\rho_2 \in \mathcal{D}_2} \langle\psi_1|\rho_2|\psi_1\rangle} \right\rceil. \quad (4)$$

- Theorem 1 and Theorem 2 give us that the QSVP error exponent is

$$\lim_{n \rightarrow \infty} \frac{-\ln P_{e,\min}(p, \{|\psi\rangle\langle\psi|^{\otimes n}\}, \mathcal{D}_{2,n})}{n} = -\ln \sup_{\rho_2 \in \mathcal{D}_2} \langle\psi|\rho_2|\psi\rangle,$$

which recovers the composite Chernoff exponent at $s = 0$.

- Fully composite setting: $|\mathcal{D}_1|, |\mathcal{D}_2| > 1$.

Theorem 3: Upper bounds (General)

Let $\mathcal{D}_1, \mathcal{D}_2$ be finite compact sets of d -dimensional quantum states. Then, for $p, \delta \in (0, 1)$ the following bound holds:

$$n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) \leq \left\lceil \frac{\ln\left(\frac{\sqrt{|\mathcal{D}_1||\mathcal{D}_2|p(1-p)}}{\delta}\right)}{-\ln F_{\max}} \right\rceil. \quad (5)$$

Let $\mathcal{D}_1, \mathcal{D}_2$ be compact sets of d -dimensional quantum states, possibly of infinite cardinality, satisfying $F_{\max} \leq c$ for some $c \in (0, 1)$. Then, for $p, \delta \in (0, 1)$ the following bound holds:

$$n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) \leq \left\lceil \frac{2 \ln\left(\frac{\sqrt{p(1-p)} K_{c,d}}{\delta}\right)}{-\ln F_{\max}} \right\rceil, \quad (6)$$

where $K_{c,d} := \binom{N+d^2-1}{N} c^{N/2}$ and $N := \max\{1, \lceil \frac{d^2\sqrt{c}-1}{1-\sqrt{c}} \rceil\}$.

Extension: Differentially Private QHT

- Private QHT: Each state passes through an ε -locally differentially private quantum (ε -LDPQ) channel before testing.

Definition of an ε -LDPQ Channel

A quantum channel $\mathcal{M}$ is ε -LDPQ, for $\varepsilon \geq 0$, if

$$\sup_{\rho, \sigma} E_{e^\varepsilon}(\mathcal{M}(\rho) \| \mathcal{M}(\sigma)) = 0,$$

where $E_\gamma(\rho \| \sigma) := \text{Tr}[(\rho - \gamma\sigma)_+]$ is the quantum hockey-stick divergence [2], and we denote the set of ε -LDPQ channels by LDP_ε .

- Given some $\mathcal{M} \in \text{LDP}_\varepsilon$, we denote the locally private sets as

$$\mathcal{D}_i^\mathcal{M} := \{\mathcal{M}(\rho_i) : \rho_i \in \mathcal{D}_i\}, \text{ for } i = 1, 2.$$

- Define the (ε, δ) -LDPQ sample complexity as

$$n_\varepsilon^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) := \inf_{\mathcal{M} \in \text{LDP}_\varepsilon} n^*(p, \mathcal{D}_1^\mathcal{M}, \mathcal{D}_2^\mathcal{M}, \delta),$$

and the minimal quantum hockey-stick divergence as

$$E_{1,\min} := \inf_{\rho_1 \in \mathcal{D}_1, \rho_2 \in \mathcal{D}_2} E_1(\rho_1 \| \rho_2).$$

Theorem 4: Sample Complexity of Private QHT

Let $\mathcal{D}_1, \mathcal{D}_2$ be compact sets of d -dimensional quantum states, possibly of infinite cardinality, satisfying $F_{\max} \leq c$ for some $c \in (0, 1)$. Then, for $\varepsilon \geq 0$ and $p, \delta \in (0, 1)$, we have

$$\frac{e^\varepsilon}{(e^\varepsilon - 1)^2} \frac{(1 - \delta/p)^2}{E_{1,\min}^2} \leq n_\varepsilon^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) \leq \left\lceil \left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right)^2 \frac{4 \ln(K_{c,d} \sqrt{p(1-p)}/\delta)}{E_{1,\min}^2} \right\rceil. \quad (7)$$

Summary and Future Work

- Comparison of our composite bounds with the known simple results:

	Simple [1]	Composite	Private simple [2] $p = \frac{1}{2}, \delta = 0.1$	Private composite $p = \frac{1}{2}, \delta = 0.1$
Lower bound	$\frac{\ln\left(\frac{p(1-p)}{\delta}\right)}{-2 \ln F(\rho_1, \rho_2)}$	$\frac{\ln\left(\frac{p(1-p)}{\delta}\right)}{-2 \ln F_{\max}}$	$\frac{16 e^\varepsilon}{25 (e^\varepsilon - 1)^2 E_1(\rho_1 \ \rho_2)^2}$	$\frac{16 e^\varepsilon}{25 (e^\varepsilon - 1)^2 E_{1,\min}^2}$
Upper bound	$\left\lceil \inf_{s \in [0,1]} \frac{\ln\left(\frac{p^s(1-p)^{1-s}}{\delta}\right)}{-\ln \text{Tr}(\rho_1^s \rho_2^{1-s})} \right\rceil$	$\left\lceil \frac{2 \ln\left(\frac{\sqrt{p(1-p)} K_{c,d}}{\delta}\right)}{-\ln F_{\max}} \right\rceil$	$\left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right)^2 \frac{2 \ln 5}{E_1(\rho_1 \ \rho_2)^2}$	$\left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right)^2 \frac{4 \ln(5 K_{c,d})}{E_{1,\min}^2}$

- Future work: asymmetric, M -ary, and non-identical composite QHT.

References

- H.-C. Cheng et al., "An Invitation to the Sample Complexity of Quantum Hypothesis Testing," *npj Quantum Information*, 2025, arXiv:2403.17868.
- H.-C. Cheng et al., "Sample Complexity of Locally Differentially Private Quantum Hypothesis Testing," arXiv:2406.18658, 2024.